

```

import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for numerical operations
import matplotlib.pyplot as plt # Import Matplotlib for plotting

# Step 1: Create a synthetic image representing a clear scene
img = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array
cv2.circle(img, (50, 50), 30, 200, -1) # Draw a filled or outlined circle on the image

# Step 2: Simulate low light by drastically reducing contrast and brightness
low_light = cv2.convertScaleAbs(img, alpha=0.2, beta=0) # Scale and shift pixel values

# Step 3: Apply Histogram Equalization to mathematically stretch the contrast
enhanced = cv2.equalizeHist(low_light) # Enhance contrast using histogram equalization

# Step 4: Display the final enhanced image
plt.figure(figsize=(8,4)) # Create a new figure window for display
plt.subplot(121), plt.imshow(low_light, cmap='gray'), plt.title('Low Light Image')
plt.subplot(122), plt.imshow(enhanced, cmap='gray'), plt.title('Enhanced via Equalization')
plt.show() # Show the final plot window to the user

```

